Total no. of Pages....

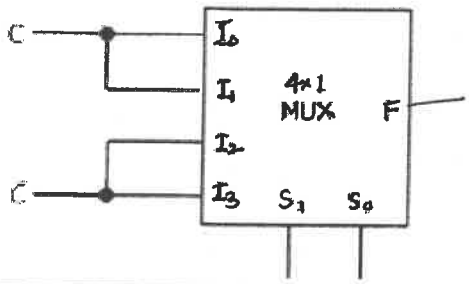
Roll no. .....

Degree: B.Tech (EVDT) Semester: 3rd – Course work
MID-SEMESTER EXAMINATION, SEP 2026

Course Title: Digital Circuits & Systems
Course Code: VTECC305

Duration: 1:30 Hours
Max. Marks: 20

Note: Attempt all five questions. Missing data/information (if any) may be suitably assumed and mentioned in the answer.

1.	(a)	Implement a XNOR gate using NAND gate only.	**2 Marks**	**CO1**
	(b)	A 4 × 1 Multiplexer is shown in the Figure below. Determine the minimized expression for output F. 	**2 Marks**	**CO2**
2.	(a)	Implement the following function using 8:1 multiplexer F (A, B, C, D) = $\sum m$ (1,2,5,6,8,9,13,14,15) Assume B, C, and D as the selection variables.	**2 Marks**	**CO2**
	(b)	Minimize the following functions using K-map. i. W(a,b,c,d)= Σ (2,12,13) ii. X(a,b,c,d)= Σ (7,8,9,10,11,12,13,14,15) iii. Y(a,b,c,d)= Σ (0,2,3,4,5,6,7,8,10,11,15) iv. Z(a,b,c,d)= Σ (1,2,8,12,13)	**2 Marks**	**CO2**
3.	(a)	11001, 1001 and 111001 correspond to the 2's complement representation of which numbers?	**2 Marks**	**CO1**
	(b)	Implement a full adder using decoder.	**2 Marks**	**CO2**
4.	(a)	Design a combinational circuit to multiply two unsigned 2-bit binary numbers A_1A_0 and B_1B_0. Obtain the truth table, derive the Boolean expressions, and draw the logic circuit.	**2 Marks**	**CO2**
	(b)	Make a NOR gate circuit diagram using CMOS logic family.	**2 Marks**	**CO2**
5.	(a)	Realize $y = AB + CD$ using minimum number of NAND gates only.	**2 Marks**	**CO1**
	(b)	Perform the subtraction of the given numbers: i. $(100010.001)_2 - (10110.1)_2$ using 2's complement. ii. $(1631.1899)_{10} - (745.17)_{10}$ using 10's complement	**2 Marks**	**CO1**